

RAGMAC Sample Report

Analysis of Document Indexing Performance

Executive Summary

This report evaluates the performance characteristics of the ragmac indexing pipeline across six document formats. Tests were conducted on a MacBook Pro with M3 Pro chip and 36GB RAM running macOS 15.2.

Key Findings

1. Native NLEmbedding processed 10,000 chunks in 4.2 seconds.
2. CoreML BGE-small model processed the same workload in 12.8 seconds.
3. PDF conversion via PDFKit extracted text with 97.3% accuracy.
4. EPUB documents with complex formatting showed 94.1% text extraction.
5. DOCX files with embedded tables preserved text content correctly.

Chunking Quality

The chunker targets 2048 characters per chunk with 10% overlap. Paragraph boundaries are preferred split points, falling back to sentence boundaries and finally hard character splits when necessary. In testing, 89% of chunks aligned with natural paragraph breaks, preserving semantic coherence.